Forces & Equilibrium

ANSWER KEY

Section 1: Resolving Forces in Two Dimensions

1 A force of magnitude 50 N acts at an angle of 30° above the horizontal. Find its horizontal and vertical components.

a. **Horizontal:** 43.3 N**, Vertical:** 25.0 N ✓

b. Horizontal: 25.0 N, Vertical: 43.3 N

c. Horizontal: 35.4 N, Vertical: 35.4 N

d. Horizontal: 50.0 N, Vertical: 0.0 N

$F_x = 50\cos(30^\circ) = 50(\sqrt{3}/2) \approx 43.3$ N. $F_y = 50\sin(30^\circ) = 50(1/2) = 25.0$ N.

2 Two forces $F_1 = [8,\ -3]$ N and $F_2 = [-5, 7]$ N act on a particle. What is the magnitude of the resultant force?

a. 5 N ✓

b. 7 N

c. $\sqrt{41}$ N

d. 12 N

Resultant $\mathbf{R} = F_1 + F_2 = [8 - 5,\ -3 + 7] = [3, 4]$ N. Magnitude $|\mathbf{R}| = \sqrt{3^2 + 4^2} = 5$ N.

3 A particle is held in equilibrium by three coplanar forces: P acting East, Q acting North, and a force of 13 N acting at an angle θ South of West where $\tan\theta = 5/12$. Find the magnitude of P.

a. 12 N ✓

b. 5 N

c. 13 N

d. 8 N

For equilibrium, horizontal components balance: $P = 13\cos\theta$. Since $\tan\theta = 5/12$, $\cos\theta = 12/13$, so $P = 13(12/13) = 12$ N.

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Section 2: Friction & Limiting Equilibrium

4 A block of mass 6 kg rests on a rough horizontal table where the coefficient of friction is $\mu = 0.35$. Taking $g = 10$ m/s^2, what is the maximum frictional force that can act on the block?

a. 21 N ✓

b. 60 N

c. 18 N

d. 24 N

Normal reaction $R = mg = 6(10) = 60$ N. Maximum static friction $F_{\max} = \mu R = 0.35(60) = 21$ N.

5 A horizontal force of 15 N is applied to the 6 kg block in the previous question (where $F_{\max} = 21$ N). What is the actual frictional force acting on the block?

a. 15 N ✓

b. 21 N

c. 0 N

d. 6 N

Since the applied force (15 N) is less than the limiting friction (21 N), the block remains at rest and friction balances the applied force: $F = 15$ N.

6 A particle of mass 4 kg is placed on a rough plane inclined at 30° to the horizontal. If the particle is on the point of slipping down the plane, find the coefficient of friction μ.

a. $\mu = \tan(30^\circ) = \frac{\sqrt{3}}{3} \approx 0.577$ ✓

b. $\mu = \sin(30^\circ) = 0.5$

c. $\mu = \cos(30^\circ) \approx 0.866$

d. $\mu = 0.25$

At limiting equilibrium on an incline, $\mu = \tan\alpha$. For $\alpha = 30^\circ$, $\mu = \tan(30^\circ) = 1/\sqrt{3} = \sqrt{3}/3 \approx 0.577$.

Section 3: Inclined Planes & Equilibrium

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7 A crate of mass 10 kg is held in equilibrium on a smooth slope inclined at 30° to the horizontal by a force P acting parallel to the slope. Taking $g = 10$ m/s^2, find P.

a. 50 N ✓

b. 86.6 N

c. 100 N

d. 25 N

Resolving parallel to the slope: $P = mg\sin(30^\circ) = 10(10)(0.5) = 50$ N.

8 For the crate in the previous question, what is the normal contact force exerted by the slope on the crate?

a. 86.6 N **(**$50\sqrt{3}$ N**)** ✓

b. 50 N

c. 100 N

d. 75 N

Resolving perpendicular to the slope: $R = mg\cos(30^\circ) = 100(\sqrt{3}/2) = 50\sqrt{3} \approx 86.6$ N.

9 A body of mass 8 kg rests in limiting equilibrium on an inclined plane. The normal reaction is 64 N. Taking $g = 10$ m/s^2, find the angle of inclination α of the plane.

a. $\alpha = \arccos(0.8) \approx 36.9^\circ$ ✓

b. $\alpha = 30^\circ$

c. $\alpha = 45^\circ$

d. $\alpha = 60^\circ$

$R = mg\cos\alpha \Rightarrow 64 = 8(10)\cos\alpha \Rightarrow 80\cos\alpha = 64 \Rightarrow \cos\alpha = 0.8 \Rightarrow \alpha \approx 36.9^\circ$.

Section 4: Lami's Theorem & Connected Systems

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10 A lamp of weight $W = 60$ N is suspended by two light strings making angles of 30° and 60° with the horizontal ceiling. Find the tension in the steeper string.

a. $30\sqrt{3}$ N ≈ 52.0 N ✓

b. 30 N

c. 60 N

d. 45 N

The angles with the vertical are 60° and 30°. The two strings are perpendicular ($30^\circ + 60^\circ = 90^\circ$). Resolving along the steeper string: $T_2 = W\cos(30^\circ) = 60(\sqrt{3}/2) = 30\sqrt{3}$ N ≈ 52.0 N.

11 A light inextensible string passes over a smooth peg. A mass of 5 kg hangs on one side and a mass of 3 kg hangs on the other. What downward external force must be applied to the 3 kg mass to maintain equilibrium? ($g = 10$ m/s^2)

a. 20 N ✓

b. 50 N

c. 30 N

d. 10 N

Tension in string is $T = 5g = 50$ N. On the 3 kg side: $T = 3g + F \Rightarrow 50 = 30 + F \Rightarrow F = 20$ N.

12 A ring of mass m is threaded on a rough horizontal wire with coefficient of friction $\mu = 0.4$. A light string attached to the ring is pulled with tension T at an angle of 45° above the wire. Find the value of T (in terms of mg) when the ring is in limiting equilibrium.

a. $T = \dfrac{0.4mg}{\cos(45^\circ) + 0.4\sin(45^\circ)} = \dfrac{0.4mg}{1.4\frac{\sqrt{2}}{2}} = \dfrac{4\sqrt{2}}{7}mg$ ✓

b. $\dfrac{\sqrt{2}}{2}mg$

c. $0.4mg$

d. mg

Resolving vertically: $R + T\sin(45^\circ) = mg \Rightarrow R = mg - T\sin(45^\circ)$. Horizontal friction: $F = \mu R = 0.4(mg - T\sin(45^\circ))$. For limiting equilibrium: $T\cos(45^\circ) = 0.4(mg - T\sin(45^\circ)) \Rightarrow T(\cos 45^\circ + 0.4\sin 45^\circ) = 0.4mg \Rightarrow T = \frac{0.4mg}{1.4(\sqrt{2}/2)} = \frac{4\sqrt{2}}{7}mg$.

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